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Simulation of street landscape design based on machine learning and entertainment design robots: An interactive entertainment design experience

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ABSTRACT

Traditional design methods usually present design ideas through plans and models, but this method can not truly simulate the design effect. Therefore, this research aims to use computer game simulation and virtual robotics technology to provide a more interactive and realistic design experience. The study creates a virtual street environment with elements such as streets, buildings and vegetation through computer game simulation. Participants can move and interact in this environment through virtual robots, experience the process of street landscape design, adjust the street layout, add landscape elements, and observe the design effect in real time according to their own design intentions. Compared with traditional design methods, the experimental results show that the design experience using computer game simulation and virtual robot technology is more vivid, creative and interactive. Participants can better perceive the design effect and adjust the design scheme in real time. This interactive entertainment design experience method can help designers and participants better understand and evaluate the design scheme, and improve the design effect and participation.

1. Introduction

Street landscape design plays an important role in urban planning and architectural design. Traditional design methods usually rely on plane plans and static models, which can not accurately simulate the real effect of the design scheme. In the traditional design method, plane plan and static model can not give participants real perception and interactive experience, and it is difficult for designers and participants to accurately evaluate and adjust the design scheme. Traditional methods usually require a lot of time and resources to make models and implement tests, which limits the innovation and exploration of design solutions. Therefore, it is of great significance to explore more interactive and authentic design experience methods to promote the development of street landscape design. The rise of computer game simulation and virtual robot technology has brought new possibilities to street landscape design. Computer game simulation techniques can create virtual urban environments in which elements such as streets, buildings, vegetation, etc. are simulated. Virtual robot technology can realize the interaction and operation of participants in the virtual environment. By combining these two technologies, you can create an interactive entertainment design experience that allows participants to perceive and adjust design effects in real time. The application of 3D design for street landscapes has brought new opportunities and challenges to urban street landscapes through advanced technologies such as the Internet of Things, virtual reality, and interactive robots [1]. The application of virtual reality technology provides designers with more intuitive and realistic design tools [2]. Designers can use virtual reality technology to create realistic street landscape models, design and modify them in a virtual environment. This realistic simulation allows designers to better showcase their creativity and ideas, and communicate more effectively with clients or decision-makers. Virtual reality technology can also help designers evaluate the effectiveness of different design schemes and conduct spatial layout and streamline analysis, thereby improving the quality and efficiency of design [3]. The introduction of interactive robots has added more fun and interactivity to street landscape design [4]. Designers can integrate interactive robots into street landscapes, making them a bridge between citizens and design works. For example, designers can create art installations with human–computer interaction capabilities, allowing citizens to interact with them through touch, sound, or gestures. This interactivity can enhance citizens' sense of participation and belonging to the street landscape, making design no longer limited to static forms, but a medium for direct interaction and communication with people.

The simulation method of street landscape design based on computer

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game simulation and virtual robot can provide more real, vivid and creative design process through virtual environment and interactive design experience. Participants can move, interact and adjust the design scheme in the simulated urban environment through virtual robots, observe the design effect in real time, and quickly iterate. Therefore, the simulation method of street landscape design based on computer game simulation and virtual robot can make up for the shortcomings of traditional design methods and provide an interactive entertainment design experience. This method is expected to promote the innovation and development of street landscape design, and bring more accurate and effective design schemes to the field of urban planning and architectural design. This article combines advanced technologies such as the Internet of Things, virtual reality, and interactive robots to design street landscapes, which can better understand the needs and preferences of urban residents. By collecting and analyzing data from IoT sensors, designers can obtain information about resident behavior and preferences, which helps optimize street landscape design based on their actual needs, providing an environment that is closer to their living habits and aesthetic appreciation. Designers can use virtual reality technology to simulate the effects of different schemes during the design phase, including the layout of street landscapes, material selection, color matching, etc. This can reduce trial and error and modification in the actual construction process, thereby reducing waste and unnecessary costs [5]. By introducing sensors and networking devices into landscape facilities, real-time monitoring of facility operation status, energy consumption, maintenance needs, and other information can be achieved, allowing designers and urban managers to better understand the operation of the facilities. Robots can provide navigation, directions, and consultation services for residents on the streets, as well as personalized recommendations and suggestions based on their needs [6]. This can not only improve the overall management level and service quality of the city, but also enhance the satisfaction and happiness of residents. Therefore, through the application of technologies such as the Internet of Things, virtual reality, and interactive robots, designers can create more intelligent, intuitive, and interactive street landscape works, providing better living experiences for urban residents. This not only improves the urban environment, but also brings more development opportunities for the cultural and creative industry of the city.

2. Related work

The literature indicates that traffic warning robot systems based on the Internet of Things have unparalleled cost-effectiveness advantages [7]. The design of the system includes key aspects such as robot body design, mobile app design and virtual operation structure design, network design, and robot autonomous tracking and walking design. The literature has achieved higher performance and reliability by optimizing the appearance and function of robots. Robots are made of lightweight materials, with compact dimensions and flexible movements to adapt to various complex traffic environments. The robot is equipped with high-definition cameras and sensors such as LiDAR, which can accurately perceive the surrounding traffic situation and display corresponding traffic warning information as needed. The literature provides a user-friendly mobile application that allows users to conveniently monitor and control traffic warning robots remotely [8]. Through the APP, users can obtain real-time traffic information, set traffic warning information, and interact and control with robots through virtual operation structures. This design enables users to make timely decisions and adjustments, improving the flexibility and operational efficiency of the system. The literature has achieved seamless communication and data exchange between robots and servers through Internet of Things technology. In this way, traffic warning robots can obtain real-time traffic conditions and other related information, and update traffic warning information in a timely manner to provide accurate warnings and guidance [9]. The literature utilizes modern navigation and positioning technologies to enable robots to autonomously track and walk without human intervention [10]. Robots can perceive surrounding obstacles and traffic conditions in real-time through sensing devices such as LiDAR and visual sensors based on preset paths and targets, and make corresponding navigation and walking decisions. In the literature, the application layer is the top layer of the entire platform, with IoT services as the core [11]. This layer is responsible for interacting with users, receiving their instructions and needs, and converting them into control commands for robots. Through IoT technology, the application layer can exchange and communicate data with other devices and systems, achieving connection and interaction with the external environment [12]. The ROS robot operating system provides rich functional modules and algorithm libraries for tasks such as data processing, perception, and decision-making. In the literature, the underlying control layer is implemented by an embedded system, including various sensors and actuators, to perceive the surrounding environment of the robot and execute specific motion control instructions [13]. The design of the underlying control layer emphasizes real-time and reliability, and achieves precise control of the robot through interaction with the information decision-making and processing layer. The literature platform is centered around the ROS robot operating system, which can fully utilize the existing functional modules and algorithm libraries of ROS to accelerate the speed of system development and deployment [14]. The design of the entire platform also takes into account cost and reusability, enabling the control of mobile robots on a low-cost basis. The literature sorts and screens indicators based on the characteristics of the data, in order to obtain more effective evaluation indicators. By statistical analysis of street landscape related data, the weight and importance of each indicator can be determined, and a comprehensive evaluation result can be obtained. The application of the street landscape evaluation index system in literature can provide scientific basis for street planning and design [15]. By evaluating various aspects of the street landscape, we can understand the advantages and disadvantages of the street environment, and make targeted improvements and optimizations. These evaluation indicators can also serve as the basis for evaluating the quality of the street landscape, helping decision-makers and planners make more reasonable decisions and design choices.

3. Design of interactive robots based on the Internet of Things and virtual reality

3.1. IoT communication solution

Wi Fi is a short distance wireless communication technology widely used in office and home environments. It has strong mobility and can achieve wireless connection of devices within a local area network. Considering the communication methods between mobile phones and robots, after comprehensively comparing the above wireless communication methods, it has been decided to use Wi Fi as the communication method. Wi Fi has a small delay and can meet the national standard distance requirements for triangular warning signs used in motor vehicles. In open environments such as highways, Wi Fi signals are less affected by interference and have better communication stability. Wi Fi also has good transmission distance, which can meet the communication needs between robots and mobile phones.

Through computer game simulation technology, create a virtual urban environment, including streets, buildings, vegetation and other elements. This simulated environment can accurately reflect the real world, and has a high degree of freedom of editing and manipulation. Designers and participants can conduct real-time design operations in a virtual environment, adjusting and arranging various elements to create the ideal streetscape. Virtual robotics allows participants to become the controllers of virtual robots in a simulated environment to experience the design process in an immersive manner. Through the combination of computer game simulation, virtual robot and Wi-Fi communication technology, we will achieve a novel simulation experience of street landscape design. Participants can move freely through the simulated urban environment through virtual robots, observe and perceive the design effects, and make instant adjustments and improvements to the street landscape as needed. This interactive entertainment design experience will bring greater engagement and creativity to streetscape design, fostering design innovation and exploration.

3.2. Control model design

This article uses a two wheel differential robot, and the relationship between the change in angle and the distance traveled by the left and right wheels of the robot can be expressed by formula (1).

$$d\theta = \lim_{\Delta t \to 0} \frac{ds2 - ds1}{L} \tag{1}$$

For classical PID control, it can be divided into two forms: positional PID and incremental PID. The expression of positional PID is shown in formula (2).

$$u(k) = K_P e(k) + K_I \sum_{i=0} e(i) + K_D [e(k) - e(k-1)] \tag{2}$$

The expression of incremental PID is shown in formula (3):

$$\begin{gathered} \Delta u(k) = u(k) - u(k-1) = u(k) = \\ K_P[e(k) - e(k-1)] + K_I e(k) + K_d[e(k) - 2e(k-1) + e(k-2)] \end{gathered} \tag{3}$$

Proportional Integrated Derivative (PID) control is a common controller design method used to adjust the output of the system to approach the set target value. It calculates the control variables based on the deviation between the current position and the target position of the system, including proportional, integral, and differential terms. However, traditional positional PID control has some problems. Due to the accumulation of error integrals, each control output is affected by all past states, which may lead to an increase in accumulated errors. In addition, the overall computational load is also relatively large. To address these issues, incremental PID control has been proposed. As shown in Fig. 1, in the incremental PID, the output of the controller is no longer a single position error, but the position difference calculated at two adjacent sampling times. This can eliminate the association with the previous system error, only related to the previous and previous two errors, and avoid the accumulation of errors.

In Fig. 1, the core program of the lower computer software can update the current speed and steering parameters of the robot, so that the robot's motion state can be adjusted based on the new speed and steering parameters. It is possible to reset the CCR comparison value for PWM and limit the upper and lower limits of CCR. By preset the comparison value of CCR, the speed and direction of the motor can be controlled. Limiting the upper and lower limits of CCR can ensure that the motor operates within a reasonable range. When an abnormal situation occurs, the program can forcibly shut down the movement of the motor, which can ensure that the robot can safely stop running in case of problems. In the timer interrupt, this program can perform PI speed correction.

According to the operating mode, the program can receive joystick commands or perform autonomous tracking. By receiving different instructions, different motion modes can be achieved, and the program can also collect information on the robot's current actual speed and cumulative displacement, which can be used to monitor and evaluate the robot's operating status. In order to ensure timely control of the robot's motion status, the update speed of each state variable should be as fast as possible. According to formula (4), the interruption time of the timer can be calculated.

$$T = \frac{(ARR + 1)(PSC + 1)}{PCLK2} \tag{4}$$

Updating the speed and steering parameters of the robot needs to be achieved through a PI speed correction program. Firstly, by reading the pulse count values collected by timers TIM2 and TIM3, the encoded value of wheel rotation in each cycle is obtained, thereby obtaining the current speed information of the robot. Then, based on the expected speed and steering parameters, the speed is adjusted using the PI speed correction algorithm to calculate the expected comparison value CCR of the four channels in timer TIM1, and update them in real-time.

To ensure the safety of the robot, a main switch control is added to the motor. If excessive discharge or low battery level is found in the aircraft model battery, it may cause the robot to lose control, which is very dangerous on the road. Therefore, we set a voltage threshold detection. When the voltage is below the set threshold (such as 7 V) or when the main switch is turned off, we will force the PWM duty cycle to 0 to stop the motor from rotating and achieve an emergency stop function. The main switch can be set by triggering both the body button and the APP button, and the motor is turned off by default when the robot is turned on.

It is also necessary to collect the current actual speed and cumulative displacement of the robot, and send these status information to the OLED display screen and APP. The current driving speed of the robot will be continuously updated, and the calculated results for each cycle will replace the results of the previous cycle. The total distance traveled is recorded in a cumulative manner, and the distance traveled in each cycle is added to the results of previous cycles in order to record the cumulative distance traveled by the robot. If it is necessary to reset the driving distance to zero, it can be achieved by pressing the reset button.

Derive the voltage magnitude of the battery through simple resistance voltage division. In the STM32 chip, the pins are powered by 3.3 V, and the maximum voltage that the AD collected pins can tolerate is 3.3 V. In order to obtain more accurate voltage values, a 12 bit AD converter was used. According to formula (5), calculate the voltage of the battery using the converted numerical value.

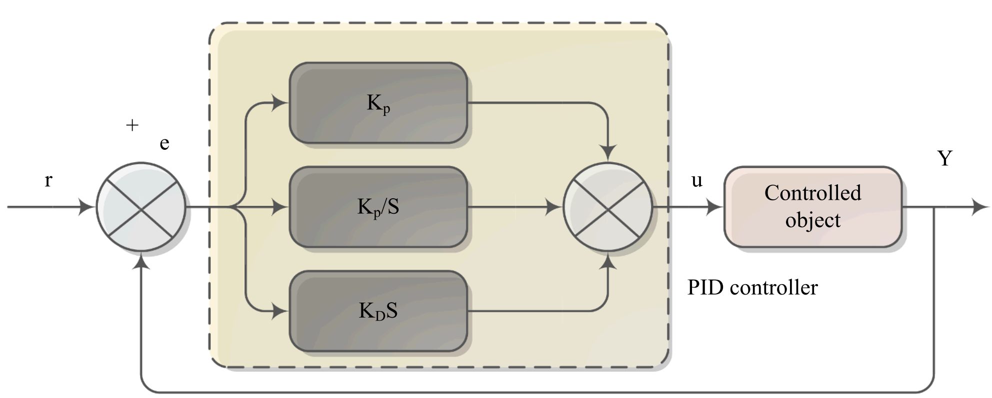

Fig. 1. Incremental PID control structure diagram.

$$\text{Volt} = \frac{\text{digital} \times 11 \times 3.3}{2^{12}}(\text{V}) \tag{5}$$

In order to reduce measurement errors, a timer TIM1 was used to set the interrupt to 5 ms. This indicates that voltage values were continuously sampled ten times within ten cycles, and the average of these ten values was taken as the estimated battery voltage. The entire process takes 50 ms, which means updating the sampling value of the battery voltage every 50 ms, which can improve the stability and accuracy of the data.

3.3. Error correction

According to the measurement principle of acceleration, in order to perform error correction, this article adopts the method of ellipsoidal fitting. The ellipsoidal fitting method can approximate the distribution of actual measured values by fitting an ellipsoidal model. In specific implementation, a series of rotational data are collected, and then an ellipsoid is fitted through a mathematical model, so that the ellipsoid can match the distribution of actual measurement points as closely as possible. Through ellipsoidal fitting, a calibration model can be obtained to correct the measurement errors of accelerometers or compasses, thereby improving the accuracy of the data.

Establish a numerical model to describe the error distribution of accelerometer or compass measurements. Assuming that the error model can be represented as an ellipsoid, its mathematical form is shown in formula (6):

$$\begin{cases} x_e = (x_a + d_x)\cdot g_x \\ y_e = (y_a + d_y)\cdot g_y \\ z_e = (z_a + d_z)\cdot g_z \end{cases} \tag{6}$$

Among them, E represents the error, (x, y, z) represents the measurement value, and (a, b, c) represents the center point of the ellipsoid. The goal is to find the optimal parameter values (a, b, c, A, B, C) through ellipsoidal fitting method, so that the corrected values are approximately distributed on a sphere. The calculation of a sphere can be done using formula (7):

$$x^2 + y^2 + z^2 = R^2 \tag{7}$$

The process of ellipsoidal fitting involves collecting a series of rotational data, then fitting the parameters of the ellipsoid through a mathematical model, and then calculating the corrected values. The ultimate goal is to distribute the corrected values on a sphere, thereby reducing measurement errors. The corrected values are substituted into the formula of the sphere and compared with the theoretical radius of the sphere (i.e. the square of the gravitational acceleration g) to construct the variance u, as shown in formula (8):

$$u = x_e^2 + y_e^2 + z_e^2 - g^2 \tag{8}$$

Among them, u represents variance, and g represents gravitational acceleration. Replace the correction value with the measured value to obtain the representation of an ellipsoid, as shown in formula (9):

$$\begin{aligned} u = {} & g_x^2 x_a^2 + d_x^2 g_x^2 + 2x_a d_x g_x^2 + g_y^2 y_a^2 + d_y^2 g_y^2 + 2y_a d_y g_y^2 + g_z^2 z_a^2 + d_z^2 g_z^2 \\ & + 2z_a d_z g_z^2 - g^2 \end{aligned} \tag{9}$$

For the convenience of calculation, make some assumptions. Assuming that the center point of the ellipsoid is consistent with the average of the measured values, i.e.:

$$V = \begin{bmatrix} x_a^2 & y_a^2 & z_a^2 & x_a & y_a & z_a & 1 \end{bmatrix}^T \tag{10}$$

$$P = \begin{bmatrix} a & b & c & d & e & f & g \end{bmatrix}^T \tag{11}$$

$$\begin{cases} a = g_x^2 \\ b = g_y^2 \\ c = g_z^2 \\ d = 2d_x g_x^2 \\ e = 2d_y g_y^2 \\ f = 2d_z g_z^2 \\ g = d_x^2 g_x^2 + d_y^2 g_y^2 + d_z^2 g_z^2 - g^2 \end{cases} \tag{12}$$

Substitute V and P into the variance formula (13) to obtain the following form:

$$u = V^T \cdot P \tag{13}$$

To solve the parameter P, perform partial derivative solving on formula (13). The partial derivative of u over P can be expressed as formula (14):

$$\begin{cases} \frac{\partial U}{\partial a} = \sum 2u\frac{\partial u}{\partial a} = \sum 2ux_a^2 = 0 \\ \frac{\partial U}{\partial b} = \sum 2u\frac{\partial u}{\partial b} = \sum 2uy_a^2 = 0 \\ \frac{\partial U}{\partial c} = \sum 2u\frac{\partial u}{\partial c} = \sum 2uz_a^2 = 0 \\ \frac{\partial U}{\partial d} = \sum 2u\frac{\partial u}{\partial d} = \sum 2ux_a = 0 \\ \frac{\partial U}{\partial e} = \sum 2u\frac{\partial u}{\partial e} = \sum 2uy_a = 0 \\ \frac{\partial U}{\partial f} = \sum 2u\frac{\partial u}{\partial f} = \sum 2uz_a = 0 \\ \frac{\partial U}{\partial g} = \sum 2u\frac{\partial u}{\partial g} = \sum 2u = 0 \end{cases} \tag{14}$$

Further simplify and write formula (14) in the form of formula (15):

$$\sum V\cdot u = \sum V\cdot(V^T\cdot P) = \sum V\cdot V^T\cdot P = \sum B\cdot P = 0 \tag{15}$$

For each attitude, acceleration measurements on the x-axis, y-axis, and z-axis were provided, and the variance value u^2 was also calculated. The results are shown in Table 1.

Import the collected data from the six points into the calibration algorithm formula described earlier, and solve for each calibration parameter. The results are shown in Table 2.

3.4. Control effect analysis

In the simulation of street landscape design based on computer game simulation and virtual robot, the choice of control system is very important for the regulation of analog voltage and the input voltage of robot drive module. In order to achieve speed control and control the torque of the motor, we use two different control systems. During the simulation, two control systems were used to control the input voltage of the robot drive module. The goal is to achieve speed control by adjusting

Table 1
Measurement values of the six position accelerometer before calibration.

axial position	x-axis	y-axis	z-axis	u^2
1	0.26625	0.29291	9.66831	−3.06228
2	−9.50193	0.34702	2.55520	−0.34732
3	4.72011	−7.49837	3.95437	−0.80259
4	9.78259	−0.25638	0.08382	0.02554
5	−0.19453	−9.99126	−0.17359	2.39677
6	0.05530	−0.01438	−10.06446	5.55907

Table 2
Calibration parameter table.

a	−10.38522	x_a	0.01237
b	−9.77572	y_a	−0.16495
C	−10.36684	Z_a	0.21955
d	−0.26090	g_x	0.98913
e	3.23504	g_y	0.97331
f	−4.51118	g_z	1.00126
g	1000	C	−10.40438

the simulated voltage, and then control the torque of the motor. In some cases, PID control systems can provide good stability and dynamic response, but parameters need to be carefully adjusted to meet specific control requirements. Fuzzy control systems can handle complex nonlinear systems and provide good control performance in situations where fuzzy and fuzzy rules are relatively clear. However, the parameter adjustment of fuzzy control systems is relatively complex and requires expert experience or optimization methods. By comparing the performance of the two control systems in Fig. 2, it is possible to evaluate their performance differences in controlling the input voltage of the robot drive module.

The results in Fig. 2 indicate that the fuzzy PID control system has higher accuracy, smaller overshoot, and shorter adjustment time compared to traditional PID control systems. These advantages make the fuzzy PID control system widely used in control systems and better meet practical needs.

When the system enters a stable state, the two control algorithms add the same disturbance signal, as shown in Fig. 3.

By observing Fig. 3, it can be seen that the fuzzy PID control system can recover to a stable state more quickly when dealing with disturbances, and there are fewer fluctuations and oscillations generated during the process.

4. Robot image recognition and processing

4.1. Street landscape image processing

When performing geometric projection processing on panoramic street view images, formulas (16) and (17) can be used to project them into horizontal perspective images.

$$X_p = \frac{W_i}{2\pi} - \frac{\sqrt{3}}{2} * \frac{W_i}{2\pi}\tan\left(\frac{\pi}{6} - \frac{2x_i\pi}{W_i}\right) \tag{16}$$

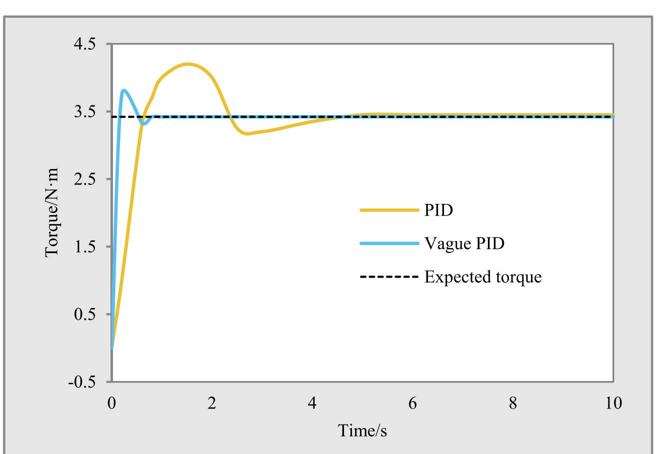

Fig. 2. Comparison chart of control system response curves.

$$y_p = \frac{W_i}{2\pi} - \frac{\sqrt{3}}{2} * \frac{W_i}{2\pi}\frac{\tan\left(\frac{\pi}{6} - \frac{y_i - \frac{H_i}{2}}{W_i}\right)}{\cos\left(\frac{\pi}{6} - \frac{2x_i\pi}{W_i}\right)} \tag{17}$$

For the cropping and projection of street view images, formulas (18) and (19) can be used:

$$x_c = \frac{W_i}{2} - f\tan\frac{\pi}{6} \tag{18}$$

$$y_c = \frac{H_i}{2} - f\tan\frac{\pi}{6} \tag{19}$$

A commonly used method can be used when projecting street view panoramic images into fisheye images. The fisheye image essentially displays the panoramic image of the street view in a directional projection on a plane tangent to the hemisphere. To complete this process, create a circular image that is equivalent in size to a hemisphere and connect each pixel of the street view panoramic image to each pixel in the fisheye image. The projection process can be achieved by constructing the relationship between pixel coordinates (x_f, y_f) in fisheye images and pixel coordinates (x_i, y_i) in panoramic images. The projection formula is as follows:

$$x_f = \frac{W_i}{2\pi} - \frac{W_i}{2\pi}\cos\frac{2x_i\pi}{W_i}\cos\left[\left(\frac{1}{2} - \frac{y_i}{H_i}\right)\pi\right] \tag{20}$$

$$y_f = \frac{W_i}{2\pi} - \frac{W_i}{2\pi}\cos\frac{2x_i\pi}{W_i}\sin\frac{2x_i\pi}{W_i} \tag{21}$$

For the merged projection of street view images into fisheye images, formulas (22) and (23) can be used:

$$x_f = R - \frac{R*\left(\frac{W_i}{2} - x_i\right)}{f_s} \tag{22}$$

$$y_f = \begin{cases} \frac{f*R}{f_s} - R & (\text{when vertical angle } = 0) \\ R + \frac{R*\sqrt{{f_h}^2 - \left(\frac{W_i}{2} - x_i\right)^2}}{f_s} & (\text{when vertical angle } = 45) \\ R - \frac{R*\left(\frac{H_i}{2} - y_i\right)}{f_s} & (\text{when vertical angle } = 90) \end{cases} \tag{23}$$

Through this projection method, street view images can be combined into images with a fisheye effect, allowing observers to observe the street view scene from a broader perspective and obtain more comprehensive information. This fisheye image projection helps to provide a more realistic and vivid visual experience, and can be applied in fields such as virtual reality and computer vision.

4.2. Analysis of street view elements

In the design and planning of commercial pedestrian streets, the landscape elements within the street range are intertwined and complex, possessing unique expressive abilities and conveying rich information to people. The road surface of the street not only carries traffic and guides pedestrian flow, but also has a richer and more active visual expression ability. The pavement of commercial pedestrian streets can be composed of different materials, colors, and forms to enhance the diversity of the landscape. In line with the colors and materials of the buildings along the street, various materials such as stone, ceramic tiles, glass

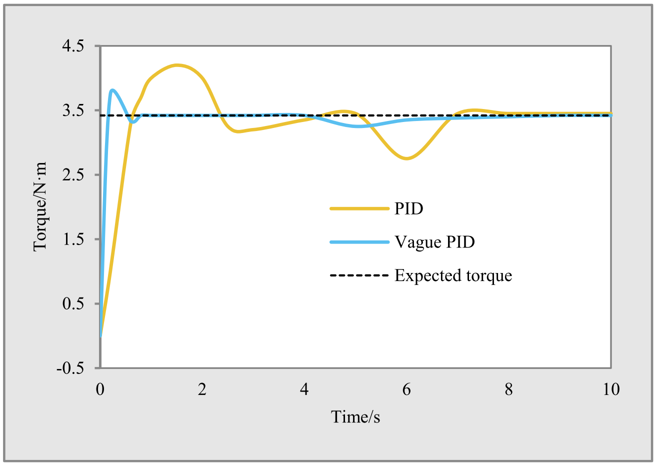

Fig. 3. Comparison diagram of control systems under disturbance signals.

decoration, or metal can be selected for the road surface to showcase the historical and cultural characteristics of the street. Some commercial pedestrian streets still have copper reliefs interspersed between stone paving, combined with elements such as ground spotlights and light strips, forming a diverse paving effect, enhancing the texture and affinity of the road surface.

As an interactive entertainment design experience, street-wide landscape elements need to be interwoven and complex to express a unique visual language and convey a wealth of information. In the computer game simulation, by simulating the regulation of voltage and the input voltage of the robot drive module, the virtual robot can simulate the real pedestrian flow and interact with the participants. They can make corresponding actions according to the behavior and instructions of the participants, increasing the sense of entertainment and participation. Computer game simulations can provide a variety of options when designing street surfaces. The walkways of commercial pedestrian streets can be made of different materials, colors and forms to enhance the diversity of the landscape. Participants can experience the feel and effect of different materials through the game simulation, including stone, tile, glass decoration, metal and more. Such simulations can help designers better show the historical and cultural characteristics of the street and optimize them according to the preferences of the participants. In the virtual simulation of the commercial pedestrian street, the texture and affinity of the pavement can be enhanced by embellishing the stone pavement with copper relief, setting ground spotlights and lamp strips. Participants can experience the effects of these landscape elements through computer games, including the change of light and shadow, the touch of texture, etc., so as to enhance the immersion and realism of the design experience.

The skyline forms a visual pathway to the sky and is a way to experience the visual transparency of the street. The ground floor and some second floors of buildings along the street are used as dense interfaces for commercial activities, with large openings facing the street to attract pedestrians and maximize the display of commercial content. Similarly, some opaque spaces are designed as boutique display windows, becoming a means of attracting pedestrians. In terms of volume and area, storefronts and display windows are not particularly prominent, but they can maximize the attractiveness of commercial activities, such as brand, trend, product temperament, and discounted price information.

The purpose of advertising is for commercial promotion and display, but if it is set too densely or placed improperly, it can not only lead to landscape fragmentation and chaos, but also cause serious visual obstruction, affect the skyline of buildings, and produce dazzling lighting problems, which have a negative impact on the visual quality of the landscape. When considering the visual evaluation of street landscapes, it is also necessary to consider the influence of street furniture. Street furniture includes pedestrian seats, telephone booths, newsstands, indicator light boxes, public water dispensers, mailboxes, roadside sculptures and planting facilities. These furniture are mainly practical and can serve as supplements and embellishments to the landscape environment. Most commercial pedestrian streets use linear arrangement of furniture to meet the needs of tourist flow and usage. But if the position is set improperly or the volume is too large, it will block the line of sight of pedestrians and affect the efficiency of finding targets. Therefore, when visually evaluating the street landscape, it is necessary to comprehensively consider the layout and planning of advertising and furniture to ensure visual transparency and efficiency. Proper placement and size control can reduce visual interference and improve the quality of street landscape.

4.3. Skyline optimization strategy

Through computer game simulations, experiencers can enter a virtual urban environment and interact with virtual robots. These virtual robots can perform various actions and behaviors according to the instructions of the experiencer, such as shuttling between buildings and simulating different angles of observation. This interactive experience will allow visitors to more intuitively feel the impact of streetscape design on the skyline. Through a combination of computer game simulations and virtual robots, the experiencer is free to explore the streets of the virtual city and change and adjust elements of the skyline through interaction with the virtual robots. Users can adjust the appearance of the skyline by changing the height, shape, material and other parameters of the building. At the same time, through the guidance and suggestions of virtual robots, the experiencer can also try different design concepts and layouts to achieve a more beautiful, unique or creative skyline effect. This kind of street landscape design simulation based on computer game simulation and virtual robot can not only provide a flexible and interactive design platform for designers, but also bring a participatory design experience for experience-takers. Through their own creativity and imagination, the experiencer can work with the virtual robot to create a unique street landscape. This interactive entertainment design experience will give the experiencer a deeper understanding and appreciation of the beauty of the skyline, while also deepening the understanding and knowledge of street landscape design. Through such design experience, people can more intuitively feel the importance of design to urban skyline, and then improve the level and quality of urban landscape design.

The skyline connects the buildings, terrain, and sky in the enclosed space. The skyline is composed of concrete buildings and an empty sky, creating a strong visual contrast, which makes it an important element to attract the attention of observers and create a deep impression. The architectural fa ç ade of urban streets is the main way to perceive the urban skyline. Through the street side fa ç ade of urban streets, people can appreciate the changes in the skyline from different perspectives. The skyline is composed of numerous variables, and each beautiful skyline contains many internal and external aesthetic elements. People usually pay attention to the rhythm of the skyline, which shows the staggered changes in the height, width, and length of the physical space through the changes of "real empty real empty". By considering these factors comprehensively, people can better appreciate and evaluate the beauty of the skyline.

The boundary line between the top of a building and the sky is usually composed of line segments. When multiple buildings are arranged together, they form a skyline composed of multiple line segments, which are continuous but have abrupt changes and exhibit relatively discrete characteristics. The variation of the contour line at the top of a building is actually a special case of the undulation of the landscape range. The changes in the contour lines at the top of the building also reflect the richness of the scenic space and scenery. When the form, height, and layout of buildings become more diverse, the changes in line segments on the skyline become more diverse, bringing people a more interesting and rich visual experience.

The changes in the alignment of the street skyline have an impact on the visual experience of pedestrians due to the aspect ratio of the street section. According to existing research, when the viewing angle is above 45°, pedestrians tend to focus more on perceiving the details of space and buildings; When the viewing angle is around 18°, the vision begins to become scattered, and pedestrians begin to pay attention to the relationship between buildings and the surrounding environment. The skyline outlined by buildings along the street becomes a significant boundary to distinguish between virtual and real spaces; When the viewing angle drops below 14°, the space becomes more open, and buildings become important elements protruding from the background. At this point, the importance of contour lines becomes more prominent.

In the field of mathematical statistics, the commonly used indicators for measuring changes in discrete values are standard deviation or standard deviation. When describing the skyline of a street, if there is a significant height difference between the top contour segments of a single building, the resulting skyline will have significant undulations and a large standard deviation. On the contrary, if the height difference is small, the standard deviation of the skyline will be smaller.

Suppose there is a set of data $x_1, x_2, \ldots, x_n$ (all real numbers), where the average value is:

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i i = 1, 2, \cdots n \tag{24}$$

The standard deviation of the data can be calculated using formula (25):

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2} \tag{25}$$

By calculating the average of the sum of squared differences and taking its square root, the standard deviation of the sample data is obtained. The larger the standard deviation, the greater the difference between the data and the greater the fluctuation of the skyline. A smaller standard deviation indicates a smaller difference between the data and a relatively flat skyline. When introducing the concept and calculation method of standard deviation into the skyline variation index, it is necessary to consider the influence of line segment length on the results. To eliminate the potential impact of pixel scale differences on different images in digital images, the formula can be normalized by using formula (26) to consider the influence of line segment length:

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\Delta l_i \cdot \Delta h_i)^2} i = 1, 2, \cdots n \tag{26}$$

The normalized standard deviation is obtained by dividing each difference term by the length of the line segment for normalization. But when considering the influence of line segment length, the product S of horizontal and vertical scales, i.e. area, should also be considered. Therefore, use formula (27) to further normalize:

$$\sigma = \frac{1}{S}\sqrt{\frac{1}{n}\sum_{i=1}^{n}(\Delta l_i \cdot \Delta h_i)^2} \tag{27}$$

In this way, by dividing by the area S of the image, the influence of size is eliminated, and a more accurate skyline change index is obtained.

4.4. Evaluation of street view elements

Table 3 shows the indices of differences, colors, and variations for each street image. The difference index represents the degree of difference between images, the color index represents the color richness of the image, and the change index represents the degree of change in the image. Sky width is an index that represents the degree of openness of the sky in an image, with higher values indicating a more open sky.

Table 4 shows the values of each building in terms of width, building height, aspect ratio, and aspect ratio. Width refers to the horizontal span of a building, while building height refers to the vertical height of the building. The aspect ratio refers to the ratio between the height and width of a building, while the aspect ratio refers to the ratio between the width and height of a building.

4.5. Street view perception analysis

Table 5 can be used to evaluate the spatial quality of streets by analyzing street view image data. By comparing the average values in different regions, the variation of street spatial quality within different ring road ranges was understood.

The analysis results based on street view image data in Table 5 show that the pedestrian street index, percentage of sunshine duration, sky visibility factor, and green view rate show a gradually increasing trend from the inner ring to the outer ring. This indicates that as the loop area expands, the pedestrian friendliness, sunshine duration, sky visibility, and proportion of green landscape of the street gradually increase. The traffic vehicle index, environmental occlusion index, building visibility factor, and tree visibility factor show a gradually decreasing trend from the inner ring to the outer ring, indicating that as the distance from the city center increases, the traffic flow, degree of environmental occlusion, visibility of buildings, and distribution density of trees gradually decrease. For the analysis of standard deviation, the percentage of sunshine duration, sky visibility factor, tree visibility factor, and green visibility rate have significant standard deviations, indicating that these indicators have significant changes within different ring road ranges, and the differences in street spatial quality are more significant. The pedestrian street index and color richness have a small standard deviation, indicating that the pedestrian friendliness and color richness of the street vary relatively steadily within different ring road ranges.

As shown in Table 6, as the space expands from the inner ring to the outer ring, the standard deviation of street space quality scores in each circle of the Fifth Ring continues to increase. This indicates that the street space quality in a large area of the central urban area with complex built environments also has diversity and differences. The elements of street space quality include safety, convenience, comfort, and vitality. Within the various ring roads in the central urban area, the safety score gradually increases, the convenience score remains relatively stable, the comfort score fluctuates slightly but overall is higher, while the vitality score is relatively stable.

In street plants, different tree species can bring ornamental characteristics in different seasons, increasing the diversity of the street landscape. It can be considered to appropriately mix and plant these tree species in the street landscape to create a more diverse and colorful street landscape effect. When selecting and planting roadside trees, it is necessary to follow the principles of suitability for the location and economic rationality, to ensure the adaptability and cost-effectiveness of the tree species. It is also necessary to consider factors such as the pollution resistance, wind resistance, and growth rate of tree species to ensure the healthy growth and long-term maintenance of roadside trees. The principle of consistency in height and distinctiveness in front and back. In terms of color matching and coordination, some ground cover

Table 3
List of differences, colors, and change indices.

Number	Difference index	Color index	Change index (10^{-3})	Sky width
1	7.01	4457	4.125	0.339
2	13.91	7677	26.893	0.452
3	12.59	5965	5.474	0.189
4	9.53	1882	3.807	0.228
5	10.48	6247	3.675	0.160
6	5.45	4176	5.430	0.574
7	10.05	3505	8.387	0.200

Table 4
List of width, building height, aspect ratio, and aspect ratio.

Number	width m	Building height m	depth-width ratio	Aspect ratio
1	21.331	14.648	0.6867	1.4563
2	17.059	13.145	0.7706	1.2977
3	29.804	31.890	1.0700	0.9346
4	9.915	14.475	1.4600	0.6849
5	20.029	25.237	1.2600	0.7937
6	30.224	13.601	0.4500	2.2222
7	25.960	15.576	0.6000	1.6667

Table 5
Analysis of Measurable Element Indicators Based on Street View Images.

Number	option	Central urban area	Bicyclic	Tricyclic
1	Pedestrian Street Index	0.03119	0.02700	0.03338
2	Traffic Vehicle Index	0.14480	0.18935	0.15885
3	Environmental occlusion index	0.15299	0.21826	0.16502
4	Architectural visibility factor	0.08745	0.10490	0.09667
5	Sky visibility factor	0.59607	0.51025	0.53078
6	Tree visibility factor	0.28501	0.34113	0.34073
7	Green visual acuity	0.39609	0.36224	0.40138

Table 6
Average score of street space quality and related elements.

option	Central urban area	Bicyclic	Tricyclic
Security	0.33990	0.40646	0.35442
Convenience	0.07265	0.08078	0.07978
Comfort	0.60413	0.56638	0.60609
Vitality	0.33214	0.32748	0.34008
Street quality	0.45747	0.44055	0.45840

plants such as tricolor viola, He's phoenix, purple cloud, etc. can be selected, and colored leaf plants such as chicken claw maple, colorful mulberry, and variable leaf wood can be paired to enhance the seasonal effect of color. When configuring plants, attention should be paid to the diversity of ornamental features. You can choose plants such as Heguo taro, hydrangea, crape myrtle, and wolfberry to provide richness in ornamental features, such as flower, leaf, pole, and fruit observation. Such choices can enrich the hierarchy and ornamental value of street plant landscapes. Reasonable plant configuration can create a rich sense of spatial hierarchy and rhythm. Through the layout and combination of street plant landscapes, a unique cultural atmosphere can be displayed.

5. Conclusion

In today's digital age, the rapid development of new technologies is bringing unprecedented innovation opportunities to cities. Among them, street landscape design, as an important part of urban construction, has also shown enormous development potential under the promotion of these advanced technologies. The application of street landscape 3D design is an emerging trend that integrates technologies such as the Internet of Things, virtual reality, and interactive robots in the field of urban landscape design. This innovative approach is gradually changing our understanding and experience of the urban environment. The street landscape 3D design application designed in this article has innovated on the basis of traditional landscape design methods, utilizing Internet of Things technology to achieve intelligent interconnection between landscape facilities. Through virtual robots, users can simulate planting different kinds of trees and observe their ornamental characteristics in different seasons. At the same time, the user can also choose the appropriate tree planting according to the geographical location and economic rationality, as well as the tree's anti-pollution ability, wind resistance and growth rate. Such design simulation enables users to experience the real street landscape design process in a virtual environment, increasing the sense of entertainment and participation. In the virtual environment, users can also choose different ground cover plants and colorful leaf plants to enhance the seasonal effect of color. Through the proper arrangement of plants and color coordination, users can create a street landscape with a sense of hierarchy, rhythm and rich ornamental features. This computer game simulation and virtual robot simulation of street landscape design not only provides entertainment, but also helps users better understand and master the principles and skills of street landscape design. By participating in the simulation experience, users can practice and explore the characteristics of different tree species, plant configuration and color matching, and improve their design ability and creativity.

CRediT authorship contribution statement

Wang Lin: Writing – original draft, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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